COMMUNITY-BASED TVET PARTNERSHIPS FOR LOCAL ECONOMIC DEVELOPMENT AND SKILLS TRAINING: ASSESSING ALIGNMENT AND OUTCOMES OF THE COMPUTER SYSTEM SERVICING IN SELECTED NCR LOCAL GOVERNMENT UNITS

An Undergraduate Thesis Presented to the

Faculty of the College of Education

Rizal Technological University

Mandaluyong City

In Partial Fulfillment of the Requirements for the

Degree of Bachelor of Secondary Education

Major in Computer System Servicing

by:

Bangcuyo, Hazell Anne B.

Cagud, Kelly Sheen P.

Gallardo, Christian James

Suficiencia, Aliah

Tumaliuan, Ezekiel S.

March 2027

CHAPTER 1

INTRODUCTION

Technical-Vocational Education and Training (TVET) has been widely positioned as a critical mechanism for equipping individuals with market-relevant skills, reducing unemployment, and fostering inclusive economic growth (UNESCO, 2022), particularly as global economies shift toward automation, digitalization, and green industries (Orbeta et al., 2021). In the Philippines, this policy orientation is codified through the Philippine Qualifications Framework Act (RA 10968) and operationalized through TESDA's continuing mandate to expand access to advanced technical competencies nationwide.

Despite this institutional support, contextual evidence indicates that the relationship between TVET completion and stable, well-matched employment remains inconsistent. Local tracer studies report high overall employment rates among TESDA graduates but only moderate alignment between training received and actual job requirements (Ledda, 2026; Mariano & Tantoco, 2023). Comparable patterns are observed internationally: in Bangladesh, employability gains depend heavily on sustained industry input (Alam et al., 2024), while in Ethiopia and Sweden, outcomes are contingent on active collaboration among government, training institutions, and industry. Collectively, this evidence suggests that training alone does not guarantee

employment outcomes; rather, outcomes hinge on whether training is embedded within labor-market intelligence systems, industry partnerships, and institutional support structures (Bayudan-Dacuycuy et al.,2024; Njengele, 2024) a dynamic of particular consequence in highly urbanized settings such as the National Capital Region (NCR), where Local Government Units (LGUs) are increasingly expected to function as architects of local economic strategy rather than mere implementers of national policy.

This structural tension is reinforced by broader diagnostic findings. The World Bank (2023) reports that local TVET systems in low- and middle-income countries, including the Philippines, have been slow to modernize in anticipation of Industry 4.0 and automation-driven labor shifts (Orbeta et al., 2021), a problem compounded by persistent weaknesses in national labor-market information and data-sharing systems (Bayudan-Dacuycuy et al., 2024). International comparative evidence further indicates that the depth of industry and institutional partnership, rather than training provision alone, determines TVET's contribution to local economic development: high-performing systems in Sweden, Ethiopia, and China are characterized by sustained industry input and locally grounded cooperation (Bano, 2026; Lundberg & Nilsson, 2022), whereas weak public-private collaboration, infrastructure deficits, and poor contextualization undermine outcomes elsewhere (Masiga et al., 2025; Sutiman et al., 2023).

Within this context, the NCR presents a distinctive governance setting in which LGUs are positioned to actively shape local economic strategy. While community-based vocational interventions have demonstrated potential to improve household financial outcomes in select provincial and small-scale settings (Mendoza, 2023; Tungol et al., 2025), a clear contextual gap persists: existing literature tends to evaluate TVET either at a broad national level or as isolated community livelihood interventions, rather than as structured, multi-sectoral partnership models operating within urban local governance (Edralin & Pastrana, 2023). Few studies integrate localized skills training, multi-sector partnership structures, and urban economic development within a single analytical framework.

This gap constitutes the central rationale for the present study, which examines how selected NCR LGUs construct community-based TVET partnerships, the extent to which associated curricula align with actual urban labor-market demand, and the adequacy of the policy frameworks governing the pathway from skills acquisition to sustainable local employment. By demonstrating how technical-vocational skills can be strategically managed through localized governance mechanisms, the study is intended to offer practically relevant, evidence-based guidance to local policymakers, vocational educators, and industry partners on optimizing resource allocation, strengthening graduate industry-readiness, and advancing sustainable local economic growth.

Theoretical and Conceptual Framework

This section moves from diagnosing the problem, selecting and justifying the right theoretical lens to defining terms precisely and to show how exact theory will be measured and observed in the actual data collected from NCR LGUs.

Theoretical Framework

A recurring limitation in the vocational education literature is its emphasis on macro-level policy architecture, national qualification frameworks, centralized funding formulas, and top-down curricular mandates at the expense of local implementation dynamics. These national structures are undeniably important; they define the institutional framework within which all local actors must operate. However, national-level analysis alone cannot explain a perplexing empirical pattern: partnership outcomes differ dramatically across municipalities governed by the same national regulations. One local government unit (LGU) is able to create a responsive, industry-integrated vocational training ecosystem, whereas a neighboring municipality, operating under the same national rules, produces only paper-based arrangements that result in outdated skills and poor employment outcomes for graduates.

Because national policy remains constant across these municipalities, national-level variables cannot account for the disparity. If uniform policy produced uniform results, there would be no variation. To explain it, we must

shift our analytical focus downward, to the meso- and micro-levels of local government. Specifically, to interactions between local governments, training providers, and local business networks. Local collaboration of this type is inherently variable, politically contingent, and context-dependent: it is determined by resource constraints, competing local political priorities, labor market demand, and historical levels of trust or conflict among the actors involved. To study this local reality with any rigor, researchers need theoretical tools capable of capturing these dynamics, operationalizing them into measurable constructs, and tracing how collaborative processes unfold over time.

This study uses a dual-theoretical framework, combining Lundberg and Nilsson's (2022) two-dimensional model of local TVET collaboration with Ansell and Gash's (2008) Collaborative Governance Theory. This framework is used in the Philippine TVET system, specifically in community-based partnerships in the National Capital Region (NCR), to assess the operational benefits of establishing a centralized model for Computer Systems Servicing (CSS) training.The framework, which follows an Input-Process-Output architecture, shows how a dedicated CSS center can reduce transaction costs for major IT enterprises, secure multi-year public commitment, support iterative stakeholder trust-building, and directly align graduate competencies with corporate labor demand.

Both frameworks are insufficient for the current analytical task when used separately. A study that only used Lundberg and Nilsson's model would be able to classify partnerships with a high degree of accuracy, classifying cases along its two diagnostic dimensions. However, it would not be able to explain why two partnerships with relatively similar resource inputs follow noticeably different trajectories over time. The model would demonstrate that the results vary, but it wouldn't explain how the variation came about.

On the other hand, a study that only used Ansell and Gash's Collaborative Governance Theory would produce rich, in-depth narrative accounts of how power imbalances are negotiated within a single partnership, how trust grows, and how little victories add up. However, comparability suffers as a result of this narrative richness: Ansell and Gash's theory does not, by itself, offer a standardized metric that would enable a researcher to methodically compare results across a sizable sample of cases. Although each case would be thoroughly understood, no case could be effectively compared to another.

This trade-off is resolved when the two frameworks are combined. The model developed by Lundberg and Nilsson provides a consistent, empirically based method for categorizing and comparing partnerships among a large sample of local government units. The theory of Ansell and Gash provides explanatory depth, a process-oriented explanation of the internal dynamics that result in a partnership's place on that comparative map. When combined,

the two frameworks enable a researcher to identify a partnership within a larger context of cases and provide a mechanistic explanation for how it arrived.

This dual framework's explanatory logic is remarkably accurate when applied to the Philippine technical-vocational education system. The Philippines has a devolved system of government under the Local Government Code of 1991, wherein LGUs have significant control over local infrastructure, social service delivery, and economic development, including the connections between local industry and vocational training facilities. In the meantime, national oversight is maintained by the Technical Education and Skills Development Authority (TESDA), which establishes consistent competency standards, program registration procedures, national assessment frameworks, and centralized funding allocation formulas that are applicable to all local government units.

Cross-context evidence from sources outside the Philippines further supports this applicability. In their analysis of Tanzania's decentralized TVET reform, Karia and Chanzi (2025) discovered that giving local institutions more autonomy enhanced training responsiveness only when that autonomy was combined with ongoing local commitment; autonomy alone, without accompanying commitment, did not consistently yield better results. Since the Philippine NCR shares the same fundamental structural condition—decentralized local implementation operating under a uniform national policy

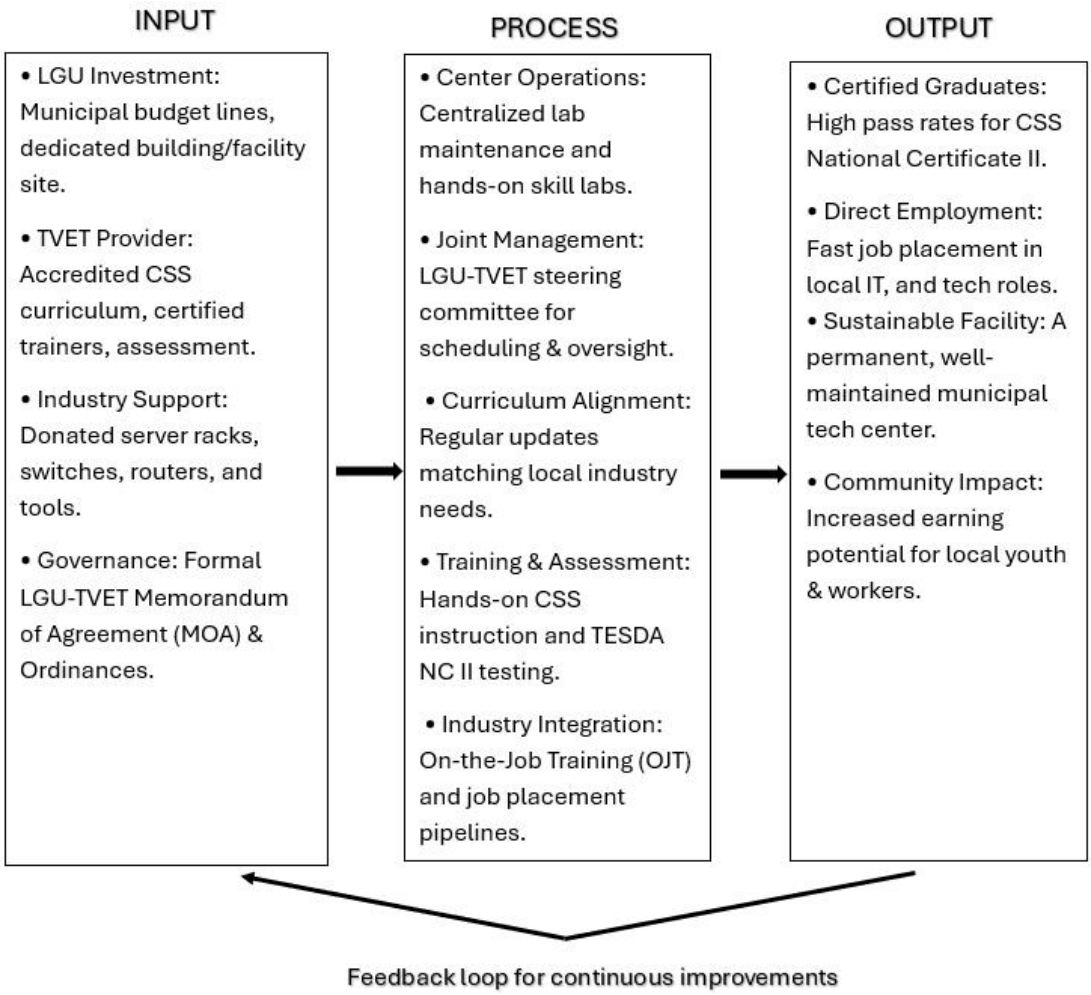
umbrella, the convergence of this finding with the Scandinavian empirical base supporting Lundberg and Nilsson's model strengthens confidence that the combined framework travels well across very different national contexts.

This is directly applicable to Computer Systems Servicing (CSS), a popular TVET program that can be seen in Metro Manila. Under TESDA's Information and Communications Technology (ICT) sector, CSS is a foundational qualification that covers fundamental skills like setting up servers, installing and configuring computer systems, and maintaining enterprise-level IT infrastructure.

The iterative phases outlined in Ansell and Gash's collaborative cycle are supported by a centralized CSS training facility, which offers a stable, impartial setting for frequent steering-committee meetings between corporate IT executives, TESDA directors, and LGU officials. This institutional and physical centralization is important because it lowers the transaction costs that would otherwise deter long-term involvement. LGU officials can focus their oversight and investment in a single, well-monitored location, and businesses don't have to coordinate across numerous, dispersed, inconsistently-run training sites. In Ansell and Gash's words, the center achieves an early "small win" by producing well-trained graduates faster thanks to this concentration of resources.

Conceptual Framework

The theoretical underpinnings and the synthesis of reviewed literature and studies, the present study conceptualizes community-based TVET partnerships using an Input–Process–Output (IPO) framework. This framework is not a novel invention of the researchers but a consolidation of the recurring relationships observed across the reviewed sources, particularly the works of Njengele and colleagues (2024), Bano (2026), Masiga and colleagues (2025), and Bayudan-Dacuycuy and colleagues (2024).



Inputs

Inputs are the resources and commitments each side brings before training begins. LGUs typically provide facility space, ordinances authorizing the partnership, counterpart funding, and utilities. TVET providers contribute curriculum standards, trainers, and accreditation requirements. Additional inputs include donated equipment, learning materials, enrollees, and the political capital embedded in memoranda of agreement between the two parties.

Processes

Processes are the mechanisms that convert these inputs into results. Center-based lab work puts facilities and equipment to direct use. Joint steering committee meetings allow LGU and TVET representatives to coordinate and resolve issues. Curriculum updates keep training aligned with industry standards, while ongoing monitoring, budget disbursement, and assessment preparation round out the operational core of the program.

Outputs

Outputs are the measurable results these processes produce: NC II certification rates, number of graduates, employment outcomes, and skills competency scores. These figures are usually what external stakeholders use to judge the partnership's success, though they only make sense when traced back to the inputs and processes that generated them.

Feedback Loop

The feedback loop connects outputs back to inputs and processes, turning the model into a cycle rather than a straight line. Low certification results can trigger curriculum revision; weak employment outcomes can prompt LGUs to reallocate resources; steering committee discussions carry lessons from one cycle into the next. This loop matters because political priorities, labor demands, and certification standards keep changing, requiring the partnership to continually adjust.

Statement of the problem

This study aimed to examine community-based Technical-Vocational Education and Training (TVET) partnerships as a mechanism for local economic development and skills training in selected local government units (LGUs) within the National Capital Region (NCR).

Specifically, this study sought to answer the following questions:

1. How do local businesses & LGU establish partnership?

2. What are the required entry level industry competencies in terms of: technical, soft skills?

3. What training program specific to BTVTED-CSS may be designed and developed?

4. How do the respondents validate the developed training program?

Scope and Limitations of the Study

Scope

This study examines how selected Local Government Units (LGUs) in Metro Manila (National Capital Region) collaborate with communities in implementing Technical Vocational Education and Training (TVET) programs, with particular focus on Computer System Servicing (CSS) as a community-based training

track, and how this program contributes to improving local employment and technical skills.

The study specifically examines whether the Computer System Servicing program is in line with the community's actual need for IT-related and technical support jobs and assesses the results it generates, including employment in computer repair or IT support roles, increased income, and the development of practical skills in hardware installation, troubleshooting, and network setup. Along with the various stakeholders involved, such as local government offices, the Technical Education and Skills Development Authority (TESDA), private computer stores and IT firms, and community organizations that host or profit from the training, it also looks into the laws and policies of the government that support or have an impact on the CSS program's implementation.

Additionally, the study highlights the advantages, disadvantages, and difficulties of providing Computer System Servicing training at the community level, especially with regard to equipment accessibility, instructor proficiency, and trainee access to real-world, hands-on learning opportunities.

Limitations

This study has several limitations. First, its scope is limited to a few selected LGUs that provide Computer System Servicing programs, so the findings may not be applicable to all areas of the National Capital Region. In

addition, due to time and resource constraints, the study uses a small sample size, so the number of trainees, instructors, and LGUs included may not fully represent the broader population of CSS program participants. The research is also hampered by limited access to records, as some LGU documents, training reports, or TESDA assessment results may be unavailable or incomplete, limiting the amount of information that can be collected.

Furthermore, there is a possibility of bias in the responses of interviewees or survey participants, such as trainees or program coordinators, who may not always provide completely accurate information, particularly if they want to present the program or their own performance favorably, or if they simply cannot recall certain technical or administrative details clearly.

This study's findings are also time-bound, reflecting only the situation at the time the research was conducted; results may change over time due to new TESDA curriculum updates, changes in LGU leadership, or shifts in the local demand for computer service skills. Furthermore, the study's success measures are limited, as it focuses on specific outcomes such as computer-related employment and National Certificate (NC) attainment, rather than capturing the full range of long-term effects that the CSS program may produce, such as career advancement or entrepreneurship in IT services.

Lastly, external factors such as the broader economy, the lingering effects of the pandemic on hands-on technical training, and political or administrative changes are not directly examined in this study, even though

they may still influence the outcomes observed in the Computer System Servicing program.

Significance of the Study

The findings of this study are beneficial to the following groups of people:

Community-Based TVET Graduates and Trainees – The observation developed from this paper aims to enhance the overall quality of local training programs, ensuring that future learners receive industry-standard skills that directly translate to sustainable employment, entrepreneurship, and improved economic mobility.

Future Researchers- This study will help the future researcher to guide them and make it as their references for better research.

Local Government Units (LGUs)- Our study will provide LGU officials, local legislators, and economic development officers with empirical data on the strengths and bottlenecks of current TVET ordinances. It will serve as a guide for crafting evidence-based local policies, optimizing budget allocations, and designing more effective community-driven livelihood programs.

Technical Education and Skills Development Authority (TESDA)- The results of this research can help TESDA provincial and district offices

evaluate how effectively decentralized, community-level training programs conform to national standards while adapting to localized urban demands.

TVET Program Coordinators and Trainers- The study will offer practical feedback to administrators and instructors regarding curriculum relevance and instructional quality. It will help them identify which specific skills are in high demand, allowing them to adjust training modules to better fit local market needs.

Definition of Terms

Community-Based TVET- Community-based Technical and Vocational Education and Training refers to skills training programs designed and implemented with active participation from the local community, government units, and other stakeholders. It aims to make training more accessible, context-specific, and responsive to the actual needs of the community and local labor market.

Curriculum Relevance- Curriculum relevance means the degree to which the content, learning activities, and competencies in a training program match the needs of learners, employers, and the labor market. A relevant curriculum prepares trainees with skills that are useful, current, and applicable to actual employment settings.

Employability- Employability is the ability of a person to obtain, maintain, and succeed in a job based on their skills, qualifications, attitudes,

and work readiness. In this study, it refers to the extent to which TVET graduates become competitive and acceptable to employers.

Graduate Employability Skills-Graduate employability skills are the soft and technical abilities that help graduates perform effectively in the workplace, such as communication, teamwork, problem-solving, adaptability, and job-specific competence. These skills are often considered essential for successful employment and long-term career growth.

Industry Partnership- Industry partnership refers to the active involvement of employers, businesses, and industry representatives in the planning, implementation, and evaluation of training programs. This collaboration helps ensure that curriculum content, training methods, and competencies remain relevant to real workplace demands.

Job-Training Alignment- Job-training alignment refers to the degree of match between the skills learned during training and the actual requirements of a job. When alignment is strong, graduates are more likely to apply what they learned effectively in the workplace and experience better employment outcomes.

Labor Market- The labor market is the environment where employers seek workers and individuals offer their skills and labor in exchange for wages or income. It reflects the demand for and supply of jobs, qualifications, and workforce competencies in a given area.

Labor-Market Intelligence- Labor-market intelligence is the collection and analysis of information about employment trends, industry needs, and workforce demands. It is used to guide planning, training priorities, and policy decisions so that education and skills development remain responsive to actual labor needs.

Local Government Unit (LGU)- an LGU refers to the local administrative body within Metro Manila (NCR) such as a city or municipal government that is responsible for managing public services and implementing development initiatives, including education and livelihood programs, within its jurisdiction. LGUs are examined in this study as key partners in the planning, funding, and implementation of community-based Computer System Servicing (CSS) TVET programs, particularly through their engagement in forging employment linkages, coordinating with TESDA and private sector partners, and integrating skills training into local development planning.

Local Economic Development (LED)- Local economic development refers to strategies and initiatives that improve the economic well-being of a specific locality by creating jobs, supporting enterprises, and strengthening local income opportunities. In this study, it emphasizes how TVET partnerships can contribute to employment generation and economic growth at the local government level.

Partnership- A partnership is a formal or informal collaboration among institutions, organizations, and stakeholders who work together toward a

common goal. In the context of this study, it refers to cooperation among LGUs, training institutions, employers, and communities to improve TVET delivery and employment outcomes.

Philippine Qualifications Framework (PQF)- The Philippine Qualifications Framework is a national policy framework that describes the different levels of qualifications in education and training in the country. It helps ensure that learning outcomes, certifications, and competencies are aligned with national standards and labor-market expectations.

Skills Anticipation- Skills anticipation is the process of predicting future workforce skills that will be needed by industries and communities. It helps training institutions and policymakers prepare learners for emerging occupations and changing economic conditions.

Skills Mismatch- Skills mismatch occurs when the competencies acquired through education or training do not fully correspond to the skills demanded by employers. This gap can lead to underemployment, difficulty in finding work, or limited job security among graduates.

Skills Training- Skills training is a structured learning process that develops practical abilities, technical competencies, and workplace-ready knowledge needed for a particular job or industry. It is intended to improve employability and help trainees meet the demands of current labor-market conditions.

Technical and Vocational Education and Training (TVET)- TVET is an educational and training system that equips learners with practical, job-related, and industry-oriented skills for employment, self-employment, and workforce participation. It focuses on preparing individuals for specific occupations through hands-on learning, technical instruction, and competency-based assessment.

TESDA - TESDA stands for the Technical Education and Skills Development Authority, the Philippine government agency responsible for managing and supervising technical-vocational education and training. It plays a central role in setting training standards, certification, and skills development programs.

Work-Based Learning- Work-based learning is a training approach that allows learners to develop skills in actual or simulated workplace settings. It strengthens practical competence by connecting classroom instruction with real industry experience.

Workplace Readiness- Workplace readiness refers to the set of behaviors, attitudes, and foundational skills that enable a graduate to function effectively in a professional setting. It includes punctuality, responsibility, adaptability, communication, and the ability to work with others.

Research Paradigm

A research paradigm provides the worldview or belief system that guides how research is conceptualized, designed, and interpreted. It reflects

how researchers see reality (ontology), how they gain knowledge (epistemology), and what methods they use (methodology).

This study adopts the Pragmatic Research Paradigm because it focuses on finding practical solutions to real-world issues involving community-based Technical-Vocational Education and Training (TVET) partnerships in selected Local Government Units (LGUs) in the National Capital Region (NCR). The pragmatic paradigm allows the researcher to combine quantitative and qualitative methods to gain a more complete understanding of the current implementation, effectiveness, and challenges of these partnerships. It emphasizes the use of different sources of evidence to develop realistic and evidence-based recommendations that can improve local skills training and economic development. This paradigm is appropriate because the study aims not only to describe the existing conditions but also to propose an enhanced policy framework that can strengthen collaboration among LGUs, industries, and TVET providers.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of localized and international literature, empirical studies, and institutional policy frameworks that establish the academic foundation for this investigation. The selected texts are systematically organized into key thematic domains: the global and national imperatives

Foreign Literature

The World Bank, International Labour Organization, and UNESCO's Joint Policy Statement on TVET Reform

The World Bank, in partnership with the International Labour Organization and UNESCO, released a joint press statement on July 12, 2023, ahead of World Youth Skills Day, accompanying the launch of the report Building Better Formal TVET Systems: Principles and Practice in Low- and Middle-Income Countries. The three institutions describe the report as the first rigorous global analysis of TVET challenges and reform options across developing economies, and its central diagnosis is that TVET systems in many low- and middle-income countries fail to match skills to labor-market needs and remain structurally unprepared for the sharp rise in enrolment demand projected over the coming decades.

The report states this mismatch within four converging global forces globalization, technological progress, demographic transformation, and climate change, each of which is reshaping the skills that labor markets require faster than many formal TVET systems can adapt. Beyond this general diagnosis, the report identifies specific institutional mechanisms that produce mismatch: many training providers default to teaching what they already know how to deliver rather than adapting programs to the needs of employers or learners, and TVET is frequently perceived as a second-choice educational pathway, a stigma that can depress both enrolment and employer willingness to hire graduates regardless of actual training quality.

On financing, the report notes that low- and middle-income countries devote a markedly smaller share of gross domestic product to TVET than high-income countries, and it recommends both mobilizing private financing and realigning existing public expenditure toward reform-linked, results-based allocation rather than simply expanding budgets under the existing provider-driven model. The report's three proposed transformations, moving from recognition to excellence through client-oriented, integrated training ecosystems; strengthening results-focused governance and financing tied to outcomes; and investing in data collection so that reform is evidence-driven, map closely onto the alignment, outcomes, and policy-framework constructs examined in the present study, and are treated here as an analytical framework rather than mere background, since each pillar can be

operationalized into indicators for assessing the selected NCR LGU partnerships.

For the present study, this joint statement is significant because it reframes skills mismatch as a problem of institutional design and partnership quality rather than simply resource scarcity a reframing that directly supports treating community-based TVET partnerships as a structural corrective mechanism, since their value lies precisely in their capacity to inject local labor-market and community input into program design in ways that centrally administered, provider-driven models cannot readily achieve.

It is also worth noting a methodological caution embedded in the report itself: its low- and middle-income country sample spans a wide range of governance and fiscal contexts, and its authors are careful to note that reform priorities differ across countries even as certain structural challenges under-resourced providers, weak signaling of quality to employers, and financing misaligned with outcomes recur widely. This caution is instructive for the present study's design: rather than assuming the report's global diagnosis applies uniformly across all NCR LGUs, the present research treats the report's framework as a set of propositions to be tested against the specific institutional and fiscal realities of the selected local government units, several of which differ meaningfully in revenue base, population density, and industrial composition despite their shared metropolitan setting. (*World Bank, 2023)*

Global Skills Partnerships: Exploring Transnational Education in TVET

Commissioned jointly by the British Council, the Edge Foundation, and the Association of Colleges, this research examines the growing but still underdeveloped role of transnational education (TNE) within the TVET sector. Transnational education the delivery of education and training across borders through partnerships, franchising, branch campuses, or joint programs is already a well-established feature of higher education internationally, but the study finds that its application to technical and vocational training has lagged despite considerable potential, particularly in contexts where local training capacity is limited or where industry standards are evolving faster than domestic institutions can adapt.

Drawing on international case studies, sector analysis, and interviews with practitioners and policymakers, the study finds that collaboration across borders and institutions can meaningfully improve access to quality skills training, strengthen the capacity of national TVET systems, and open new employment pathways for graduates. However, the researchers are careful to caution that successful transnational partnerships require intentional contextualization and genuine mutual learning between partner institutions; simply transferring curricula, delivery methods, or quality-assurance systems from one national context to another, without adapting them to local conditions, tends to produce weak or unsustainable outcomes.

This finding is relevant to the present study even though its subject matter cross-border education differs from the domestic, LGU-level partnerships examined here, because the same underlying principle applies at the local scale: a training program or curriculum transplanted wholesale from one barangay, city, or industry context into another, without adaptation to local labor-market conditions, risks producing the same kind of superficial, non-functional collaboration that the study identifies at the international level. The emphasis on contextualization over simple replication reinforces the present study's assumption that alignment must be assessed locally and specifically rather than presumed from the mere existence of a partnership agreement or memorandum of understanding. *(Edge Foundation, 2025)*

Decentralized Cooperation Between Industry and Local Government: The Swedish Vocational Training School Paradigm

Published in the Scandinavian Journal of Educational Research, this study by Lundberg and Nilsson examines decentralized cooperation between industry and local government in Sweden's vocational education and training system through a comparative case study of two partnerships between local municipalities and manufacturing companies operating industrial schools. The authors note that scholarly debate on cooperation among VET stakeholders has focused predominantly on the national level, with far less attention paid to how such cooperation is negotiated and sustained locally and regionally — an

explicit gap that the present study's own research problem directly mirrors at the NCR-LGU level.

The Swedish case is analytically significant because it examines cooperation within what the authors describe as a statist skill-formation system: Sweden's VET system remains nationally regulated and integrated into the public upper-secondary school structure, meaning that local cooperation between municipalities and industry must be actively sustained even though the overarching national system provides comparatively uniform, and at times insufficient, incentives for firms and local governments to invest in it. The study traces how manufacturing companies, which had strong historical involvement in vocational schools until the late 1960s before being displaced by the integration of VET into the comprehensive public system, reemerged as active partners and even owners of industrial schools following market reforms in the 1990s.

The study's two guiding research questions are directly analogous to the present research: first, what does local government–industry collaboration signify in terms of the depth of firm involvement and the consistency of local-government commitment; and second, what specific factors explain the similarities and differences observed in how such partnerships are established and sustained. Rather than treating cooperation as either present or absent, Lundberg and Nilsson's framework treats firm involvement and government commitment as variables that can range from passive, symbolic engagement

to deep, structurally embedded collaboration a distinction operationalized in the present study's conceptual framework, where local government commitment is defined as measurable allocation of budget, personnel, or sustained policy priority independent of any single administration's term, and firm involvement is defined as ranging from passive signatory status to active curriculum co-design.

For the present study, the Swedish paradigm supplies the primary anchor theory precisely because it operates at the correct unit of analysis, the localized government-industry partnership rather than at the national policy level or the individual-graduate level addressed by other theoretical traditions. Its explanatory logic, that local cooperation must be actively sustained within a nationally regulated system offering limited built-in incentive for doing so, maps directly onto the Philippine devolution context under the 1991 Local Government Code, in which NCR LGUs possess legal authority to design and facilitate local skills-training initiatives but must still generate the political will and institutional commitment to actually exercise that authority. *(Lundberg & Nilsson, 2022)*

Public-Private Development Partnerships in Sub-Saharan Vocational Training: Institutional Friction Points in Curriculum Co-Design

Published in the International Journal of Educational Development, this study by Wolde and Woldeamanuel examines the institutional friction points

that arise when public and private-sector actors attempt to jointly design vocational training curricula in sub-Saharan African contexts. The study's central concern is that public-private development partnerships in vocational training, while frequently promoted as a solution to skills mismatch, encounter recurring points of institutional tension differences in timelines, accountability structures, and priorities between government training bureaucracies and private industry partners that can undermine the depth and durability of curriculum co-design even when both parties enter the partnership with genuine intent to be collaborated.

This literature is relevant to the present study's process dimension, specifically the distinction between genuine curriculum co-design and mere consultation. Where much of the literature reviewed in this chapter treats industry involvement as a binary present or absent Wolde and Woldeamanuel's attention to institutional friction points supports a more granular reading: a community-based TVET partnership may formally include industry representatives in curriculum discussions while still failing to achieve substantive co-design, because underlying institutional incentives and operating rhythms between the LGU, the training provider, and the employer partner are misaligned. This caution reinforces the present study's methodological decision to assess partnership quality through documented practices and stakeholder interviews rather than relying solely on the

existence of a signed memorandum of agreement as evidence of functional collaboration. *(Wolde & Woldeamanuel, 2023)*

Engineering Social Equity through Inclusive TVET Models: Operational Indicators for Marginalized Urban Spaces

Attributed to IEEE Transactions on Education, this study by Thomas and Al-Ansi examines methods for fostering social equity through inclusive TVET models and proposes operational indicators for assessing how such models affect socio-economic marginalization within urban communities, using a qualitative policy-analysis approach. The study's central concern is that TVET programs, even when technically well-designed and aligned with labor-market demand, can still fail to produce equitable outcomes if program design does not explicitly account for the structural barriers geographic, economic, and social that determine who within a marginalized urban community can realistically access, complete, and benefit from training.

This literature is directly relevant to the present study's urban NCR context, where community-based TVET partnerships operate within a highly urbanized setting characterized by significant socio-economic disparity even within individual LGUs. The study's emphasis on operational indicators rather than purely qualitative or anecdotal equity assessments supports the present study's inclusion of perceived community economic resilience as a distinct output variable, since aggregate employment or income statistics can mask

uneven distribution of a partnership's benefits across different segments of a community. Reading alongside the World Bank–ILO–UNESCO and UNESCO strategy documents discussed above, this source reinforces a recurring theme across the foreign literature: that alignment and outcomes cannot be assessed purely in economic terms without also asking who specifically is being reached. *(Thomas & Al-Ansi, 2024)*

Local Literature

Rethinking Philippine Educational Hierarchies: Socioeconomic Returns of Vocational Qualifications versus Baccalaureate Pathways

Circulated as a working paper through the Social Science Research Network, this paper by Molina advances a policy-advocacy position arguing for prioritizing TVET over conventional four-year degree programs within Philippine education policy. The paper contends that the entrenched cultural and institutional preference for baccalaureate credentials in the Philippines has produced a persistent mismatch between the outputs of the higher-education system and the skills actually demanded by industry, resulting in underemployment among degree holders even as middle-skill positions remain difficult for employers to fill.

Molina's central claim is that restructuring educational hierarchies formally elevating vocational qualifications to parity with academic degrees in terms of

prestige, funding, and career pathways might generate more favorable socioeconomic returns for a larger share of the population than continued reliance on the traditional higher-education model. This argument is grounded less in original econometric estimation than in a critical reinterpretation of existing labor-market indicators and international comparisons, positioning the paper as advocacy intended to influence the direction of national policy rather than a purely descriptive account of outcomes.

This advocacy stance does not stand unopposed within the broader literature; other researchers cited within the same body of TVET scholarship have cautioned that wage and employment returns to vocational credentials in the Philippines have historically been mixed and, in some cases, lower than returns to university degrees, complicating any straightforward conclusion that vocational pathways are unambiguously superior. For the present study, the value of Molina's contribution lies less in settling this debate than in framing it: the paper crystallizes the competing hierarchies of educational value that the present research examines empirically at the community-partnership level, and it underscores the policy stakes attached to how socioeconomic returns to different educational pathways are measured and interpreted. *(Molina, 2021)*

Are We Ready for Future Middle-Level Skills Demands? An Assessment of the Philippine TVET Ecosystem

Published as a Philippine Institute for Development Studies (PIDS) discussion paper, this study by Orbeta, Gonzales, and Corpus approaches the TVET-versus-baccalaureate question through econometric and trend analysis rather than advocacy, assessing whether the Philippine TVET ecosystem is prepared to meet anticipated demand for middle-level skills as automation and Industry 4.0 technologies reshape the labor market. The authors examine enrollment and completion trends across TVET programs, assessment and certification data from TESDA, and indicators of technological change across industries to gauge the extent to which current training offerings correspond to emerging skill requirements.

Their analysis finds that while the TVET system has expanded considerably in scale, meaningful gaps remain between the competencies being certified and the more complex, technology-intensive skill sets that automation-affected industries increasingly require. The authors argue that without deliberate curriculum modernization, closer industry linkage, and forward-looking skills forecasting, the TVET system risks preparing graduates for occupations that are themselves vulnerable to displacement meaning that expansion in enrollment and certification numbers alone is not evidence of a functioning skills-supply system.

This finding is significant for the present study because it reframes the TVET-versus-baccalaureate debate central to Molina's (2021) advocacy piece: the relevant policy question is not simply which educational pathway is

superior in the abstract, but whether either pathway, as currently constituted, is adequately calibrated to a labor market undergoing rapid technological transformation. For NCR LGUs specifically, whose partnering industries increasingly span digitalized manufacturing, logistics, and business-process outsourcing, this finding underscores why the present study's alignment construct must examine not only whether training content matches current job titles but whether it anticipates the trajectory of technological change within local industry. *(Orbeta, Gonzales, & Corpus, 2021)*

Technical and Vocational Education and Training in the Philippines: In Retrospect and Its Future Directions Published in the Bedan Research Journal, this literature review and policy synthesis by Edralin and Pastrana traces the development, current status, and future directions of TVET in the Philippines, from its early institutional origins through its contemporary form under TESDA's stewardship. The authors document how successive policy shifts including the reorganization of vocational-training agencies and the introduction of the K-12 Basic Education Program's Technical-Vocational-Livelihood track to have progressively repositioned TVET within the broader Philippine education system as a mechanism for developing a skilled workforce, reducing unemployment, and supporting economic growth through competency-based education.

The authors situate this historical trajectory within the Philippines' commitments under Sustainable Development Goal 4 on quality education, arguing that future directions for Philippine TVET should include stronger alignment with global qualification standards, expanded access through flexible and technology-enabled delivery modes, and closer coordination between TESDA, higher-education institutions, and industry stakeholders. At the same time, the study identifies several persistent challenges affecting TVET effectiveness in the Philippines, including the mismatch between graduate skills and industry needs, limited stakeholder collaboration, and the continuous need to update training programs in response to technological advancement and shifting labor-market demand.

Edralin and Pastrana stress that stronger partnerships among government agencies, industries, educational institutions, and local communities are essential for ensuring that TVET programs remain relevant, responsive, and aligned with workforce requirements, and they recommend strengthening policy implementation, enhancing industry participation, and promoting collaborative efforts among stakeholders to improve program quality and sustainability. For the present study, this synthesis functions primarily as a historical anchor: it clarifies how the current institutional landscape whereinTVET and higher education operate as parallel but unevenly resourced systems came to take its present shape, and it identifies the policy directions against which contemporary reform proposals, including Molina's

(2021) advocacy for parity and Orbeta et al.'s (2021) readiness assessment, can be evaluated. It also directly supports the present study's emphasis on community-level partnership as the appropriate unit for examining whether these national-level policy aspirations translate into local practice. *(Edralin & Pastrana, 2023)*

Inputs to the Philippine Labor Market Information System and TESDA's Skills Anticipation and Prioritization of Skills Requirements Framework

Published as a Philippine Institute for Development Studies discussion paper and carried out with support from TESDA itself, this study addresses the infrastructural and institutional mechanisms needed to make skills planning credible in the Philippines in the first place. The authors note that skills mismatch in the Philippines has persisted since the 1970s, when the higher-education system began producing more college graduates than the economy could absorb, and they document that around one-third of Filipino college graduates are considered overeducated for their positions, while roughly two-thirds of TVET graduates are mismatched with their eventual occupations according to Asian Development Bank estimates cited in the paper.

The study is framed explicitly as an input to TESDA's own Skills Anticipation and Prioritization of Skills Requirements (SAPSR) Framework, and it examines the state of the Philippine labor-market information system (LMIS), reviewing available data sources, analytical capabilities, and

institutional arrangements to assess whether SAPSR is genuinely relevant and usable by its intended stakeholders. The authors identify persistent weaknesses in the timeliness, granularity, and inter-agency consistency of labor-market data as central constraints weaknesses that limit the ability of both TVET providers and higher-education institutions to calibrate their program offerings to actual and projected labor demand.

Among its recommendations, the study urges the Department of Labor and Employment to lead a more integrated LMIS architecture in coordination with TESDA, the Commission on Higher Education, the Department of Education, the Philippine Statistics Authority, the Department of Trade and Industry, and the National Economic and Development Authority, alongside private-sector stakeholders, arguing that involving these agencies ensures sectoral concerns are accounted for and increases commitment to sustained data collection. Notably, the authors specifically emphasize that local government units play a vital role in improving job matching and overall service delivery, and they call for investment in local capacity and data-analysis expertise as crucial to the framework's long-term success a recommendation that speaks directly to the present study's LGU-level focus.

For the present study, this discussion paper supplies an important methodological and policy caveat: any comparative assessment of community-based TVET partnerships' alignment with local labor-market needs is only as reliable as the underlying labor-market data available to the partnership's

stakeholders. Where such data remain fragmented at the national level, as the authors demonstrate they currently are, NCR LGU-level partnerships attempting to design locally responsive curricula may be operating with limited or outdated labor-market intelligence a constraint the present study's input dimension explicitly accounts for through its labor-market and skills-anticipation-data indicator. *(Bayudan-Dacuycuy, Epetia, Vargas, & Ocbina, 2024)*

Foreign Studies

Sustainable Employment for Vocational Education and Training Graduates: The Case of Future Skills Matching in Bangladesh

Published in the International Journal of Training Research and authored by researchers affiliated with the University of Dhaka and Kobe University in Japan, this mixed-method study examines sustainable employment outcomes for TVET graduates in Bangladesh through the lens of future skills matching. The authors note that Bangladesh is characterized by high demand for skilled workers, making TVET crucial for addressing existing skill gaps and stimulating economic development, and the study investigates the fundamental aspects of TVET's capacity to deliver high-quality skills-development programs aimed at increasing graduate employability.

The study's mixed-method design combines quantitative and qualitative components to thoroughly investigate the major challenges hindering TVET progress in Bangladesh, and its central conclusion is that graduate employability improves specifically when skills-matching mechanisms are tied to ongoing, active engagement with industry, rather than remaining dependent on static, periodically-revised curricula. This finding positions continuous employer engagement rather than one-time curriculum consultation as the operative mechanism connecting training design to actual employment outcomes, a conclusion that closely parallels the present study's process-dimension variable of continuous employer engagement and partnership monitoring.

For the present study, this research is significant because it provides direct foreign empirical support for one of the central claims embedded in the present study's Input-Process-Output conceptual framework: that the quality and continuity of industry engagement throughout the training cycle, rather than merely its presence at the outset, is the strongest determinant of whether training-to-employment pathways function effectively. The Bangladesh case additionally illustrates that skills mismatch and TVET underperformance are not uniquely Philippine or even uniquely Southeast Asian phenomena, but recur across developing economies wherever TVET provider institutions have not built in mechanisms for sustained industry feedback. *(Alam, Reza, Ogawa, & Ahsan, 2024)*

A Comparative Study of TVET Systems in China and Pakistan: Lessons for Pakistan's Economic Development

Published in Frontiers in Education and authored by a researcher affiliated with the School of Business at Xuzhou University of Technology in China, this study conducts a comparative analysis of TVET systems in China and Pakistan to draw lessons for Pakistan's economic development, examining the role of TVET in fostering financial and industrial development by drawing on China's TVET success to inform Pakistan's system reform. Using a qualitative document-analysis approach guided by the PRISMA framework, the study systematically analyzed forty-five documents published between 2010 and 2024 through thematic coding in NVivo software, drawing on a data-triangulated matrix of national policy documents, international organizational reports, and peer-reviewed academic literature.

The study's findings reveal that China's TVET system, supported by strong policy integration through its Vocational Education Law, sustained public investment, deep enterprise collaboration, and consistent industry linkages, has appreciably contributed to employment generation and broader economic development. By contrast, the study finds that Pakistan's TVET system continues to be constrained by a persistent structural mismatch between classroom instruction and the actual demands of contemporary industries:

industries in Pakistan remain reluctant to partner with training institutions that rely on outdated machinery and curricula that lag decades behind modern technological practice, and the resulting deficiency in industry investment generates what the authors describe as a vicious cycle without funding, industry collaboration becomes ineffective, and without industry collaboration, training remains disconnected from actual market needs.

The study concludes with specific, contextually adapted policy recommendations for Pakistan, including the development of a unified national qualifications framework, mandated public-private partnership mechanisms, and the integration of digital and green skills into vocational curricula, modeled partly on China's demonstrated approach. For the present study, this comparative case offers a clear illustration of the cumulative, self-reinforcing relationship between policy integration, sustained financing, and industry linkage on one hand, and skills mismatch, weak employability, and stalled economic development on the other to reinforcing the present study's Input-Process-Output framework's premise that inputs (policy basis, stakeholder participation, facilities) and processes (curriculum co-design, continuous engagement) jointly, rather than independently, determine whether TVET partnerships translate into meaningful economic outputs.

Bano's use of PRISMA-guided document analysis and NVivo thematic coding is also methodologically instructive for the present study's own documentary-analysis component, since it demonstrates how a systematic,

triangulated review of national policy documents, international reports, and peer-reviewed literature can be used to build a credible institutional profile of a TVET system without relying solely on primary survey data. The China-Pakistan comparison additionally underscores that the vicious-cycle dynamic the author identifies weak funding discouraging industry collaboration, which in turn keeps training disconnected from market needs is not simply a resource-allocation problem but a trust and credibility problem between training institutions and industry, a dynamic that plausibly recurs at a smaller scale between individual NCR LGUs and the private employers they seek to partner with, particularly in LGUs with limited prior track records of successful collaboration. *(Bano, 2026)*

Technical and Vocational Education and Training (TVET) Colleges and Industry: Strengthening Partnership-Building Practices

Published in the Journal of Vocational, Adult and Continuing Education and Training, this study examines how TVET colleges in South Africa can strengthen their partnerships with industry. The lead author, Tuletu Njengele, is affiliated with Lovedale TVET College in the Eastern Cape and previously oversaw partnerships at Buffalo City TVET College, while co-authors Penelope Engel-Hills and Christine Winberg are based at the Professional Education Research Institute of the Cape Peninsula University of Technology.

The researchers employed Activity Theory together with a responsive evaluation methodology to analyze partnership practices, situating the study within a broader literature on partnership management that the authors note is largely drawn from international contexts and does not always account for the specific policy and capacity constraints of the South African TVET sector.

The study's guiding research question was how TVET college-industry partnerships can be strengthened to benefit students, colleges, and industry simultaneously, with the explicit aim of identifying strategies for initiating and sustaining mutually beneficial collaboration. The findings reveal that partnerships function as ongoing co-configurations requiring continuous attention as institutional and industry needs change over time, rather than as one-time agreements that, once signed, can be assumed to remain functional indefinitely. Key indicators of a college's readiness to build successful partnerships included prior industry linkages, adequate institutional preparation, and clarity regarding the college's vision, mission, and values, alongside its demonstrated capacity to fulfil the specific requirements of industry partners.

The authors note that because partnerships with employers, industries, and communities play a role in financing, educational provision, and employment opportunities within TVET, productive partnerships carry a multiplier effect that extends beyond immediate training outcomes into broader social, economic, and personal benefits for graduates and their communities.

The study uncovered both persistent challenges in building, maintaining, and expanding these partnerships and successful practices reflecting what the authors describe as transformative agency among college staff, individuals within institutions who actively drove partnership development beyond what formal institutional processes alone would have produced.

For the present study, this South African case offers a directly applicable, empirically grounded framework for understanding what makes local institution-industry partnerships durable, which is central to the present research's examination of community-based TVET partnerships in NCR LGUs. The finding that partnerships require continuous institutional attention rather than one-time formation directly supports the present study's inclusion of continuous employer engagement and partnership monitoring as a distinct process variable, and its attention to prior industry linkages as a readiness indicator maps onto the present study's own profile question regarding years of engagement in community-based TVET partnership activities. *(Njengele, Engel-Hills, & Winberg, 2024)*

Multi-Sectoral Partnerships: Cross-Sectional Evidence from Uganda Shows Strong Readiness for Young Women's TVET Programs

Published in the International Journal of Innovative Research and Advanced Studies, this study provides empirical, quantitative evidence from Uganda on stakeholder readiness for multi-sectoral TVET partnerships

specifically targeting young women's employment outcomes, conducted within the framework of the Sustainable Inclusive Youth Employment Pathways (SIYEP) Project. Anchored theoretically in multi-stakeholder partnership theory, which conceptualizes government-nonprofit-private sector collaboration as a defined and analyzable governance arrangement, and in the social-ecological model of youth participation, the study assessed the readiness of local government agencies, private-sector organizations, and civil-society actors to engage in coordinated, cross-sectoral skills-development partnerships using a structured questionnaire covering demographic characteristics of respondents.

The study's findings indicate a notably high level of stated partnership willingness among surveyed stakeholders, 93.1 percent and a substantial existing base of prior TVET collaborations at 72.0 percent, suggesting favorable structural conditions on paper for multi-sectoral program implementation. However, the authors caution explicitly that stated willingness does not automatically translate into functional, accountable, and sustained collaboration in practice, a concern that echoes broader anxieties raised elsewhere in the literature regarding the persistent gap between partnership declarations and genuine implementation.

The study also found that community attitudes toward TVET were moderately supportive overall, with no statistically significant variation across different types of partnering organizations, suggesting that community perceptions of TVET are shaped more by broader socio-cultural context and

prior exposure to vocational training than by the specific identity or reputation of the partnering institution itself. Uganda's TVET sector was found to continue facing structural challenges including limited infrastructure, weak industry linkages, and persistent gender disparities in enrolment and completion, with young women's participation further constrained by cultural norms, household responsibilities, distance to training centers, safety concerns, and limited access to information about training opportunities.

For the present study, this is an important methodological and conceptual caution: surveys or interviews that measure LGU officials' or industry partners' stated support for TVET partnerships may substantially overstate the depth and functionality of those partnerships as they actually operate, unless triangulated against harder outcome measures such as completion rates, employment placements, or documented, active memoranda of agreement. This caution directly informs the present study's mixed-methods design, in which stated stakeholder support gathered through survey and interview instruments is explicitly triangulated against documentary analysis of policies and institutional records, rather than treated as sufficient evidence of partnership functionality on its own. *(Masiga, Were, & Nabwire, 2025)*

Local Studies

Labor Market Integration of Technical Education and Skills Development Authority (TESDA) Graduates

Published in the International Journal of Research and Innovation in Social Science, this descriptive-quantitative tracer study, supplemented by a qualitative component, involved 151 TESDA graduates in order to examine their labor-market integration. The study found that graduates generally achieved high overall integration into the labor force; however, the alignment between the skills acquired through training and the actual jobs obtained, as well as job security, was only moderate. This finding points to a persistent skills mismatch that continues to affect TESDA graduates even when overall employment figures appear favorable.

The distinction the study draws between employment rate and job-training alignment is methodologically significant for the present study, since it demonstrates empirically that these are statistically and conceptually distinct outcomes: a high proportion of graduates finding some form of employment does not indicate that community-based or institutional TVET partnerships are producing well-matched, secure employment. This directly informs the present study's decision to treat graduate employment rate and job-training alignment as separate output indicators within its Input-Process-Output framework, rather than collapsing them into a single composite employment measure, since doing so would obscure exactly the kind of moderate-alignment, high-employment pattern that Ledda's tracer study identifies among TESDA graduates nationally.

The study's qualitative component, which supplemented the survey findings among the 151 graduates, further suggests that graduates themselves are often aware of the gap between their training and their eventual job placement, even while reporting overall satisfaction with having found employment at all. This nuance is relevant to the present study's own planned qualitative interviews with LGU officials and industry partners, since it indicates that self-reported satisfaction among program stakeholders much like the self-reported partnership willingness identified by Masiga et al. (2025) in Ugand, should not be treated as a proxy for genuine alignment without being checked against more specific, job-level detail. *(Ledda, 2026)*

Assessment of Employability Skills of Technical-Vocational Education and Training (TVET) Graduates: Basis for an Enhancement Program

Published in the International Journal of Multidisciplinary: Applied Business and Education Research, this study assessed the employability skills of TVET graduates using a descriptive survey, document analysis, and interviews with 423 graduates and 45 employers. The study found a high overall employment rate among graduates, echoing Ledda's (2026) later findings, yet a skills mismatch persisted between what graduates were trained to do and what employers actually required on the job. Notably, employers surveyed valued communication skills most highly among the competencies assessed, ranking these above purely technical competencies.

This finding is significant for the present study because it demonstrates that employability gaps identified in Philippine TVET research are not confined to hard, technical skills alone but extend into workplace-readiness and soft-skill dimensions that formal technical curricula may under-emphasize. For the present study's conceptual framework, this supports treating graduate employability skills communication, teamwork, problem-solving, adaptability, and job-specific competence as a construct distinct from narrow technical job-training alignment, and it suggests that community-based TVET partnerships seeking to improve local economic development outcomes should attend not only to whether curricula match technical job requirements but whether they also build the workplace-readiness competencies that employers in the present study's selected NCR LGUs are likely to prioritize. *(Mariano & Tantoco, 2023)*

Tracer Study and Competency Assessment of Community-Based Training Graduates in Provincial Technical Institutes, 2018–2022

Conducted as an institutional research monograph for TESDA Marinduque, this study assessed the employment and training outcomes of community-based training (CBT) graduates from technical training institutes in Marinduque covering the period 2018 to 2022, using a descriptive-quantitative survey. The study observed high certification rates among CBT graduates;

however, a structural mismatch between training output and local employment opportunities persisted throughout the period studied.

This finding is particularly instructive for the present study because it demonstrates that even successful certification and program completion do not automatically translate into local employment absorption, especially in provincial or non-urbanized labor markets where the scale and diversity of local industry may be insufficient to absorb the volume or specialization of graduates being produced. While the present study's setting is the highly urbanized National Capital Region rather than a provincial context like Marinduque, Mendoza's finding remains a useful comparative caution: even within NCR, individual LGUs vary considerably in the density and diversity of local industry, meaning that community-based TVET partnerships in less industrially concentrated NCR districts may face a structurally similar risk of certifying graduates for whom sufficient local employment opportunities do not yet exist, regardless of training quality. *(Mendoza, 2023)*

Socio-Economic Impact Assessment of Localized Vocational Skills Interventions in Select Urban Poor Enclaves

Published in the International Review of Journal Management Studies, this study investigated the socio-economic impacts and household income changes associated with localized vocational skills interventions in selected urban poor enclaves, using descriptive statistics. The study found that

localized, community-driven skills training directly improved family financial baselines and contributed to household-level poverty reduction among participating families.

This finding empirically links targeted, community-based training not only to individual employability but to measurable local economic development outcomes at the household level, reinforcing the view that TVET can function as a poverty-alleviation mechanism when it is embedded within a community's specific economic context rather than delivered as a generic, one-size-fits-all program. For the present study, this is one of the clearest pieces of local evidence directly supporting the inclusion of household income improvement as a distinct output variable within the study's Input-Process-Output framework, separate from employment rate and job-training alignment, since it demonstrates that community-based training can produce measurable economic benefit at the household level even in contexts of an urban poor enclaves where formal employment absorption may otherwise be limited or precarious. *(IRJMS Report, 2022)*

Contextualizing Skills Training Effectiveness: A Profile-Based Evaluation of the ASK Livelihood Project in Barangay Talipapa, Philippines

Published in the Philippine Journal of Technical Vocational Education, this study conducted a profile-based evaluation of the Alalay sa Kabuhayan (ASK) livelihood project in Barangay Talipapa, contextualizing the

effectiveness of skills training relative to the demographic and socio-economic profile of program participants. The ASK Project was established to equip marginalized adults with market-relevant livelihood competencies and entrepreneurial skills, and by improving participant employability and independence, community-based programs of this kind actively advance several Sustainable Development Goals, including SDG 1 (no poverty), SDG 4 (quality education), SDG 8 (decent work and economic growth), and SDG 10 (reduced inequalities).

The research specifically examines how individual participant profiles, such as demographic backgrounds, influence both instructional delivery and ultimate learning outcomes, an approach grounded in the premise that understanding these learner-specific dynamics allows community-based training programs to better adapt their practical instruction and ensure that vulnerable populations receive accessible and effective pathways toward long-term productivity and financial stability. This barangay-level study illustrates how grassroots, community-based livelihood and skills-training programs can be evaluated for effectiveness at the most localized level of governance.

For the present study, this barangay-level evaluation offers a practical, directly transferable template for assessing the design and outcomes of community-based TVET initiatives implemented by NCR LGUs, since it demonstrates that program effectiveness cannot be meaningfully assessed in the abstract but must be contextualized against the specific demographic and

socio-economic composition of the community being served. This directly reinforces the present study's demographic-profile component (respondent age and gender) as a necessary contextual variable for interpreting differences in reported partnership outcomes across the selected NCR LGUs, rather than a mere descriptive formality. *(Tungol, Santos, & Reyes, 2025)*

Synthesis of the Review of Related Literature and Studies

Taken together, the foreign literature reviewed in this chapter converges on a shared diagnosis: across highly varied national contexts — from the Philippines and Bangladesh to Pakistan, South Africa, Sweden, and Uganda TVET systems consistently struggle to translate training provision into well-matched, secure, and economically meaningful employment, and the recurring explanation offered across sources is not primarily a shortage of training capacity but a shortage of functional, sustained, and genuinely co-designed partnership between training providers, government, and industry. The World Bank–ILO–UNESCO joint statement and UNESCO's own global strategy both frame this as an institutional-design problem rather than a purely resource problem, a framing echoed in concrete empirical terms by Bano's (2026) China-Pakistan comparison, where policy integration, sustained financing, and deep industry linkage jointly produced strong employment outcomes in one country while their absence produced persistent mismatch in the other.

The local Philippine literature largely mirrors this pattern while adding a distinctly national texture: Molina (2021) and Orbeta et al. (2021) frame the underlying policy debate over vocational-academic parity, Edralin and Pastrana (2023) trace how that debate has been partially, but not fully, resolved through instruments like the K-12 Tech-Voc track and the Philippine Qualifications Framework Act, and Bayudan-Dacuycuy et al. (2024) supply the empirical grounding for why even well-intentioned local alignment efforts may falter a fragmented, under-timely national labor-market information system that leaves TVET providers, including LGU-level community partnerships, without reliable data on which to calibrate training design. This is a critical structural caveat for the present study: NCR LGUs attempting to align community-based TVET programs with local industry needs are working, to some degree, within the same national data constraints that Bayudan-Dacuycuy and colleagues identify, meaning that locally observed alignment gaps may reflect this upstream data deficiency as much as local institutional shortcomings.

Across the reviewed studies specifically as opposed to policy literature a consistent empirical pattern also emerges regarding where significant relationships to employment and local economic development outcomes are, and are not, found. Employment rate alone is consistently high across nearly every tracer study reviewed, both foreign (Alam et al., 2024) and local (Ledda, 2026; Mariano & Tantoco, 2023; Mendoza, 2023), yet job-training alignment, job security, and where measured the household income and local

employment absorption are consistently found to be only moderate or structurally constrained. This divergence between employment quantity and employment quality is the single most consistent finding across the review, and it directly justifies the present study's decision to disaggregate graduate employment rate, job-training alignment, household income improvement, and perceived community economic resilience into four distinct output indicators rather than a single composite employment measure.

Finally, the studies reviewed converge on identifying continuity and depth of industry engagement not the mere existence of a partnership agreement as the operative mechanism linking inputs to favorable outputs. Njengele et al.'s (2024) South African finding that partnerships function as ongoing co-configurations requiring continuous attention, Alam et al.'s (2024) Bangladesh finding that employability improves specifically when skills-matching is tied to active, ongoing industry engagement rather than static curricula, and Lundberg and Nilsson's (2022) Swedish distinction between passive and deep firm involvement, all point toward the same conclusion that anchors the present study's theoretical framework: that the process dimension of a community-based TVET partnership curriculum co-design, continuous employer engagement, and partnership monitoring is what ultimately converts favorable inputs into measurable local economic development outcomes, and that Masiga et al.'s (2025) caution against equating stated willingness with functional collaboration makes clear why this process dimension must be

assessed through triangulated, documented evidence rather than survey responses alone. It is precisely this gap between partnerships that exist on paper and partnerships whose inputs and processes actually produce measurable outputs that the present study's Input-Process-Output framework and hypothesis testing are designed to address within selected NCR local government units.

A final thread worth drawing out across both the literature and the studies reviewed is the recurring caution against treating stated stakeholder support, signed agreements, or aggregate certification and employment figures as self-evidently meaningful without further disaggregation. Masiga et al. (2025) caution against equating stated willingness with functional collaboration; Bayudan-Dacuycuy et al. (2024) caution against treating national labor-market data as complete or timely; Ledda (2026) and Mariano and Tantoco (2023) both demonstrate that high employment figures can coexist with weak alignment; and Mendoza (2023) demonstrates that high certification rates can coexist with weak local absorption. Each of these cautions functions as a check against a superficial reading of partnership success, and collectively they justify the present study's triangulated, qualitative approach wherein combining graduate interviews, official interviews, industry-partner interviews, and documentary policy analysis as the minimum design necessary to distinguish community-based TVET partnerships that are merely documented

from those that are demonstrably functional and economically productive for the NCR local government units in which they operate.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter explains how the study will be carried out. It covers the research design, research locale, participants of the study, research instruments, validation of the instruments, data-gathering procedure, data analysis, and ethical considerations used in the study.

Research Design

This study will conduct a qualitative research design, specifically a case study approach to assess the alignment, outcomes, and policy framework of community-based Technical-Vocational Education and Training (TVET) partnerships for local economic development and skills training, with specific focus on the Computer System Servicing (CSS) National Certificate II (NC II) program implemented in selected Local Government Units (LGUs) within the National Capital Region (NCR). A qualitative case study design was deemed most appropriate because the study sought an in-depth, contextualized understanding of how CSS-based TVET partnerships are conceived, implemented, and experienced by the stakeholders directly involved, rather than the measurement or statistical generalization of outcomes across a large population.

The case study approach allowed the researcher to examine each selected LGU's community-based CSS training program as a bounded system, situated within its own local governance structures, resource constraints, and industry linkages. This design facilitated triangulation of multiple data sources including semi-structured key informant interviews, focus group discussions (FGDs), and document/policy analysis to produce a rich, holistic account of program alignment with local labor market needs, perceived and observed outcomes for trainees, and the policy frameworks that shape program delivery.

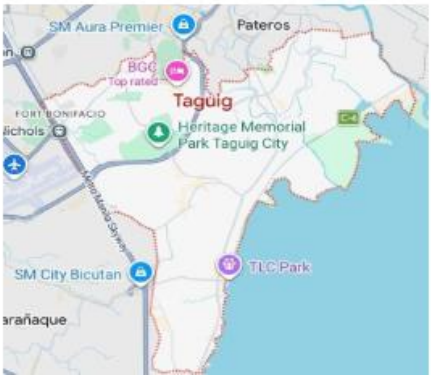
Research Locale

This study will be conducted in the cities of Makati and Taguig, two local government units (LGUs) within the National Capital Region (NCR), Philippines. These areas were purposely selected due to their active involvement in technical-vocational education and training (TVET) initiatives, as well as their different economic setups, which give the study a wider range of experiences to look into when examining community-based TVET partnerships and local economic development. This allows the researchers to look closely at how these partnerships are set up, carried out, and experienced across two NCR LGUs with different industries and local policies, giving a fuller picture of alignment, outcomes, and policy issues relevant to local economic development.

Figure 1.1 Research Locale Figure



Figure 1.2 Research locale Figure



Participants of the Study

This study will involve key people who directly connected to community-based TVET programs in Makati and Taguig, including those who help plan, run, teach, hire from, or personally went through these programs. Since this is a qualitative study, the goal is not to gather a large number of respondents, but to talk to people who can give detailed, first-hand insight into how these partnerships actually work. The participants will therefore be drawn from four main groups: LGU officials and policy implementers, who help plan and oversee the TVET partnership; TESDA representatives and trainers, who help deliver the actual training; local employers and industry partners, who hire or work with TVET graduates; and TVET graduates and trainees, who personally went through the program.

The researchers will handpick participants who are directly involved in or knowledgeable about these partnerships, rather than picking people at random. To be included, a participant must currently be connected to a TVET partnership program in Makati or Taguig, have at least one year of involvement in the program, and be willing to take part in an interview or group discussion. Interviews and group discussions will continue until the researchers notice that new conversations aren't revealing much new information anymore. Researchers will collect 20 participants to conduct the study.

Research Instruments

Since this is a qualitative study, data will mainly come from interviews and small group discussions rather than surveys, using two main tools. The first is an interview guide made up of open-ended questions prepared for LGU officials and TESDA representatives, focused on how the partnership was set up, what rules or agreements guide it, what challenges have come up, and how sustainable the program has been. The second is a group discussion guide, made up of a separate set of open-ended questions used in small group settings with TVET graduates and local employers, focused on their personal experiences with the training, the gaps they noticed, and how the training affected their work or hiring decisions.

Validation of Instruments

Before these instruments are used, they will go through a careful review process to make sure the questions are clear, relevant, and actually able to capture what the study needs.

The draft interview and group discussion guides will first be shown to three experts: someone familiar with TVET/TESDA programs, someone familiar with local governance or economic development, and the research adviser. These experts will check if the questions are clear, relevant to the research questions, and able to draw out meaningful answers, and any feedback or suggested changes will be applied to improve the guides.

After the expert review, the guides will be tried out with two to three people who are not part of the main study but share similar characteristics with the target participants, such as TVET stakeholders from a nearby LGU not included in the study. This trial run will help check if the questions are easy to understand, how long the interview takes, and if any wording needs to be clarified.

Since this study doesn't rely on statistics to prove accuracy, its trustworthiness will instead be established in other ways. The researchers will check back with a few participants to confirm that their responses were understood correctly, and will describe Makati and Taguig in enough detail that

others can judge if similar findings might apply to their own area. Proper records of how the data was gathered and analyzed will also be kept so the process can be reviewed, and the researchers will stay mindful of their own views throughout, making sure conclusions come from what participants actually said rather than personal opinion.

Data Gathering Procedure

Before starting, the researchers will get approval from their research adviser, then send formal letters to the LGUs, TESDA offices, and partner organizations in Makati and Taguig to ask permission to conduct the study.

Once approved, the researchers will coordinate with focal persons to schedule interviews and group discussions with the selected participants. Before each session, the purpose of the study will be explained clearly, and consent will be obtained before any recording begins.

Interviews and group discussions will be conducted in person or online, whichever is more convenient for the participant, and will be recorded with permission to make sure nothing important is missed. Afterward, the recordings will be transcribed word-for-word, and the researchers will go through all the transcripts, look for common ideas and patterns, and organize them into themes that help answer the research questions. Relevant documents, such as local ordinances, partnership agreements, and TESDA

guidelines, will also be reviewed and compared with what participants shared, to see if what's written on paper actually matches what happens in practice.

Data Analysis

The gathered data will be studied using thematic analysis, which involves going through everything participants said, spotting patterns or repeated ideas, and organizing them into clear themes. This process will begin with the researchers reading through all the transcripts closely to get familiar with the data, then marking down important or interesting parts and grouping similar ideas together into broader themes. These themes will then be reviewed and refined to make sure they truly reflect what participants said, given clear names and descriptions, and finally written up as findings, using participants' own words to support each theme.

Ethical Considerations

All participants involved in the research will be expected to follow the ethical standards of the study. Informed consent will be required before collecting any data, and it is important for the researchers to make sure participants understand the purpose of the research, how it will be carried out, how long it will take, and that they have the right to refuse or withdraw at any time.

Taking part in the research will be completely voluntary, and no participant will be forced or persuaded to join. No incentives will be given, so that responses remain honest and unbiased.

All personal information, such as names and contact details, will be left out of the final report, with codes or identification numbers used instead. All data gathered, including interview notes, recordings, and transcripts, will be kept secure, with only the researchers having access.

The researchers will also make sure that all information from the study is reported honestly and accurately. There will be no fabrication or falsification of findings, and all sources will be properly acknowledged and cited to avoid plagiarism. At the end of the study, the data gathered will be properly disposed of.

REFERENCES

Alam, M., Reza, S., Ogawa, K., & Ahsan, R. (2024). Sustainable employment for vocational education and training graduates: The case of future skills matching in Bangladesh. *International Journal of Training Research*.

Ansell, C., & Gash, A. (2008). Collaborative governance in theory and practice. *Journal of Public Administration Research and Theory, 18*(4), 543–571.

Bano, S. (2026). A comparative study of TVET systems in China and Pakistan: Lessons for Pakistan's economic development. *Frontiers in Education*.

Bayudan-Dacuycuy, C., Epetia, M. C., Vargas, A., & Ocbina, J. (2024). *Inputs to the Philippine labor market information system and TESDA's skills anticipation and prioritization of skills requirements framework* (Discussion Paper). Philippine Institute for Development Studies.

Edge Foundation, British Council, & Association of Colleges. (2025). *Global skills partnerships: Exploring transnational education in TVET*.

Edralin, D., & Pastrana, R. (2023). Technical and vocational education and training in the Philippines: In retrospect and its future directions. *Bedan Research Journal*.

International Review of Journal Management Studies [IRJMS]. (2022). Socio-economic impact assessment of localized vocational skills interventions in

select urban poor enclaves. *International Review of Journal Management Studies*.

Karia, A., & Chanzi, L. (2025). *[Review of decentralized TVET reform in Tanzania — full title/source not specified]*.

Ledda, R. (2026). Labor market integration of Technical Education and Skills Development Authority (TESDA) graduates. *International Journal of Research and Innovation in Social Science*.

Lundberg, E., & Nilsson, A. (2022). Decentralised cooperation between industry and local government: The Swedish vocational training school paradigm. *Scandinavian Journal of Educational Research*.

Mariano, R., & Tantoco, L. (2023). Assessment of employability skills of Technical-Vocational Education and Training (TVET) graduates: Basis for an enhancement program. *International Journal of Multidisciplinary: Applied Business and Education Research*.

Masiga, J., Were, P., & Nabwire, C. (2025). Multi-sectoral partnerships: Cross-sectional evidence from Uganda shows strong readiness for young women's TVET programs.

International Journal of Innovative Research and Advanced Studies.

Mendoza, A. (2023). *Tracer study and competency assessment of community-based training graduates in provincial technical institutes, 2018–2022* [Institutional research monograph]. TESDA Marinduque.

Molina, R. (2021). *Rethinking Philippine educational hierarchies: Socioeconomic returns of vocational qualifications versus baccalaureate pathways*. Social Science Research Network.

Narsico, D., & Narsico, J. (2024). *cited only regarding community-level economic resilience]*.

Njengele, T., Engel-Hills, P., & Winberg, C. (2024). Technical and Vocational Education and Training (TVET) colleges and industry: Strengthening partnership-building practices. *Journal of Vocational, Adult and Continuing Education and Training*.

Orbeta, A., Gonzales, K., & Corpus, J. (2021). *Are we ready for future middle-level skills demands? An assessment of the Philippine TVET ecosystem*

01

Philippine Institute for Development Studies.

Thomas, J., & Al-Ansi, M. (2024). Engineering social equity through inclusive TVET models: Operational indicators for marginalized urban spaces. *IEEE Transactions on Education*.

Tungol, M., Santos, R., & Reyes, C. (2025). Contextualizing skills training effectiveness: A profile-based evaluation of the ASK Livelihood Project in Barangay Talipapa, Philippines. *Philippine Journal of Technical Vocational Education*.

UNESCO. (2022). *[Report/title on TVET as mechanism for skills and economic growth*

Wolde, T., & Woldeamanuel, B. (2023). Public-private development partnerships in sub-Saharan vocational training: Institutional friction points in curriculum co-design.

International Journal of Educational Development.

World Bank, International Labour Organization, & UNESCO. (2023). *Building better formal TVET systems: Principles and practice in low- and middle-income countries*. World Bank.

Legal/Policy Sources (Philippine)

Republic Act No. 7796 (1994). *TESDA Act of 1994*.

Republic Act No. 10968 (2018). *Philippine Qualifications Framework Act*.